

# The simulator against our car's logs

Five comparisons this repository computes, one chart each. Each chart says what kind of evidence it is, because they are not the same kind.

## 1. The gearbox: the ratios the car ran, against the eight Toyota publishes

**INDEPENDENT CHECK** 86.7 % of moving samples within  $\pm 4\%$  of a published ratio

## 2. Boost pressure under boost: the model against the car's own boost channel

**INDEPENDENT CHECK** medians +1.9 % apart; with the old sensor reading, +25 %

## 3. Boost against air flow: the simulator's ceiling on the car's envelope

**FIT** RMS 0.135 in pressure ratio; the top bins rest on a handful of readings each

## 4. Enrichment: the mixture under boost, against the model's map

**FIT** largest cell gap 0.076 in  $\lambda$

## 5. Engine load at the steady points: the model against the ECU's load channel

**CONSISTENCY CHECK** 1.4 % with nothing fitted -- and it cannot test the breathing model

## What the three kinds mean

**INDEPENDENT CHECK** the model side came from a published source, or from a model written without this data; the logs never touched it.

**FIT** the model side was fitted to this car's logs; the chart shows how the fit sits on the data, and tests nothing.

**CONSISTENCY CHECK** the comparison cancels the model it appears to test (CLAUDE.md mistake 12).

## Not charted, and why

- Turbine housing temperature, the quantity the project protects: the car has no sensor for it.
- Knock: the model's knock integral and the car's own retard are unrelated (AUDIT.md H5). A negative result, and no script charts it yet.
- Oil temperature over a whole drive, modelled against measured: not computed yet.
- Air mass flow and coolant: equal by construction, so a chart of them would prove nothing.

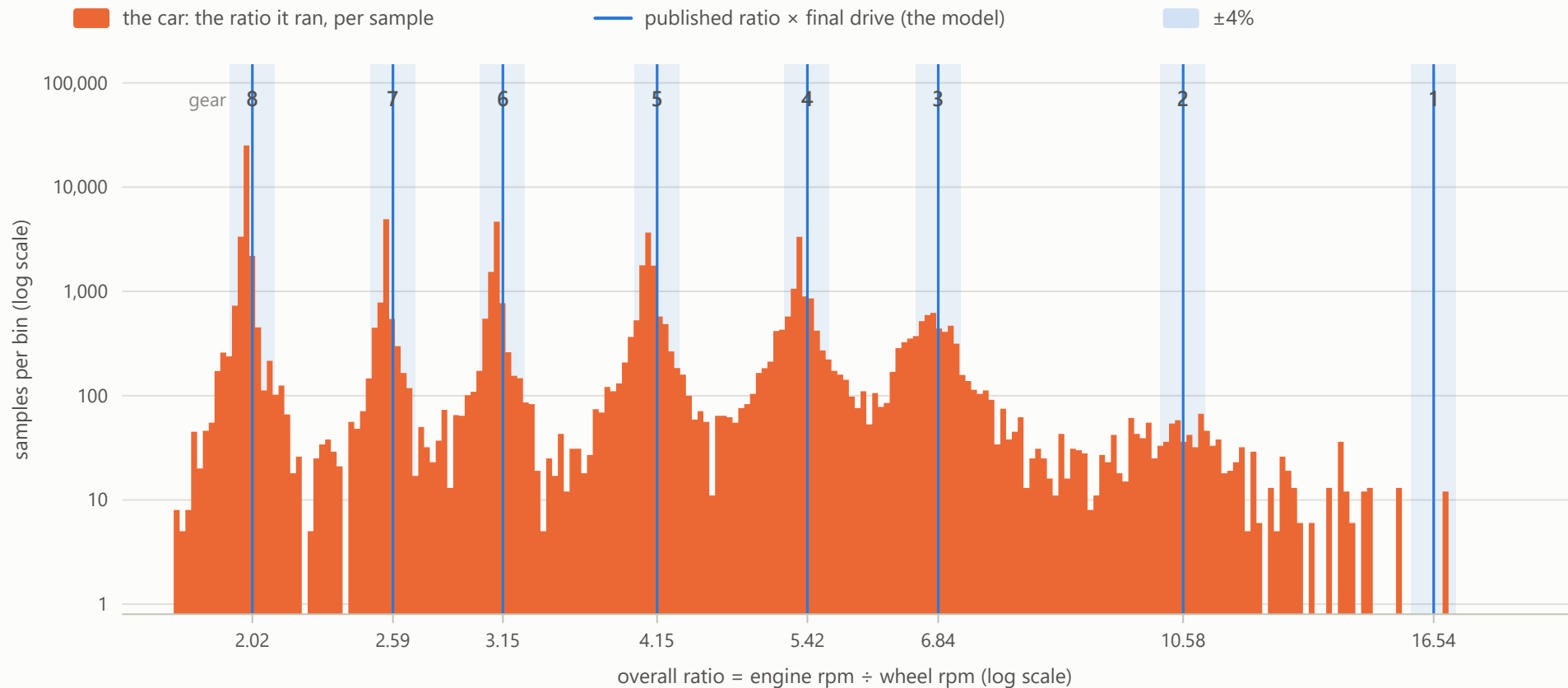
Regenerate with `python plot_sim_vs_car.py`. Every number on these pages is printed in `figures/sim_vs_car/numbers.txt`, and the tables at the end carry the values behind each chart.

# The gearbox: the ratios the car actually ran, against the eight Toyota publishes

79,134 moving samples (above 20 km/h) from 7 drives · 86.7 % lie within  $\pm 4\%$  of a published ratio

## INDEPENDENT CHECK

the ratios and the final drive are Toyota's published figures; nothing was fitted to these logs



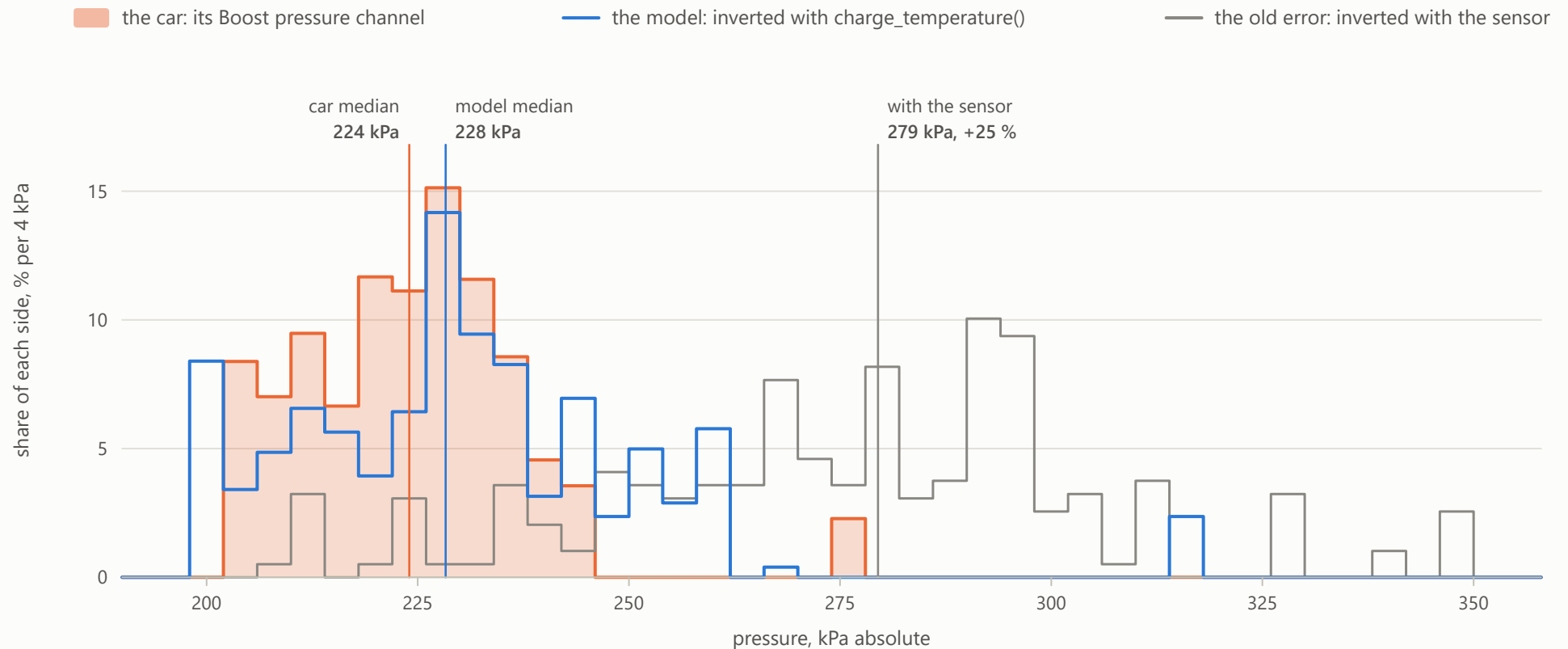
Overall ratio = engine rpm  $\div$  wheel rpm, the wheel's rpm taken from road speed and the model's wheel radius (0.33 m, whose source is not recorded). The radius shifts all eight peaks together, so the SPACING of the peaks is the independent part. Samples between the peaks are gear changes and torque-converter slip, which the model does not simulate: it assumes the converter locked. The car's own 'Actual gear' channel stops at 6, so it is not used (CLAUDE.md mistake 18). Data: data/master\_samples.csv.

## Boost pressure under boost: the model's value against the car's own boost channel

above 200 kPa · model 762 samples, car 1,097 logged rows · medians 228 and 224 kPa · gap +1.9 %

### INDEPENDENT CHECK

plant.charge\_temperature() was written without this data and has no parameter fitted to it



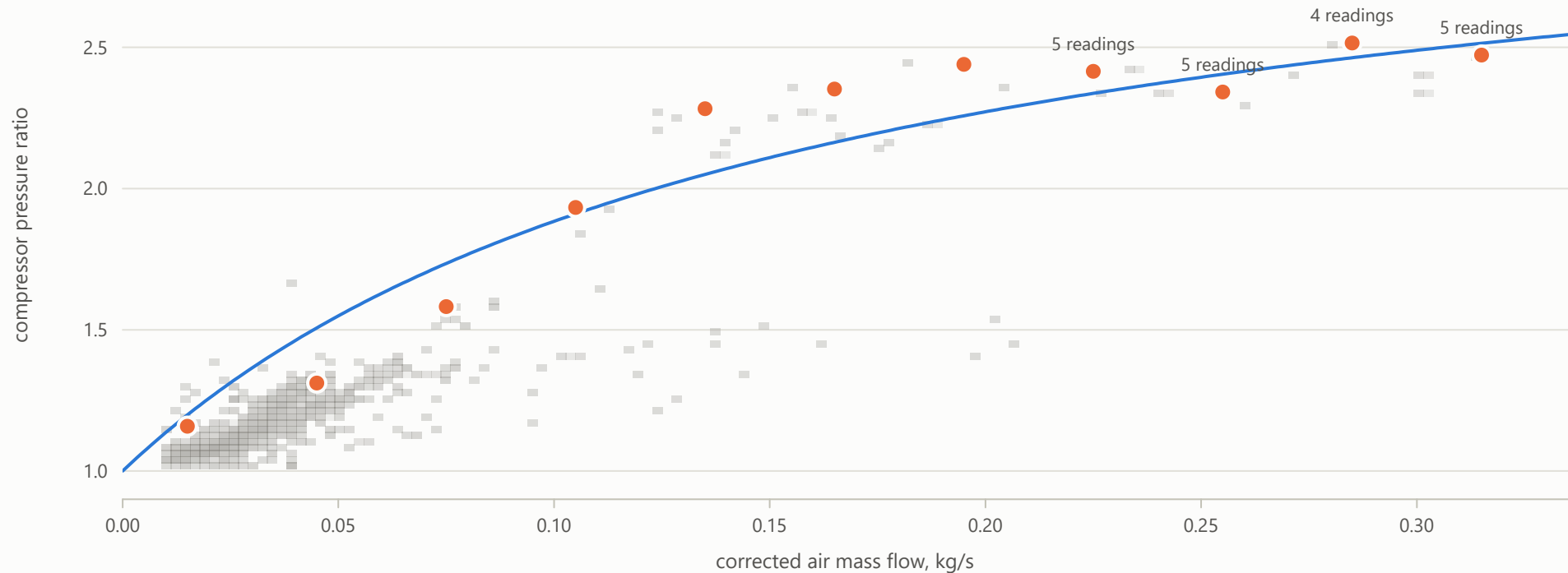
The two sides are different samples, not pairs: the logger reads one channel per row, so air mass and boost are never read at the same instant, and two populations over the same region are compared by their medians. Model: manifold pressure inverted from the measured air mass, with the modelled charge temperature. Grey: the same samples inverted with the pre-throttle sensor, which is a compressor outlet, not the charge -- the dataset's error until 10 September (CLAUDE.md mistake 13). Samples at the air-mass sensor's ceiling are excluded (mistake 7). Data: data/master\_samples.csv, logs/raw/\*.csv.

## Boost against air flow: the simulator's ceiling on the car's measured envelope

40,716 steady samples · 95th percentile per 0.03 kg/s bin · RMS gap 0.135 in pressure ratio

**FIT** the ceiling (plant.boost\_ceiling\_kpa) was fitted to this car's envelope; the chart shows the fit and tests nothing

■ the car: steady samples      ● the car: 95th percentile per bin      — the model: boost ceiling

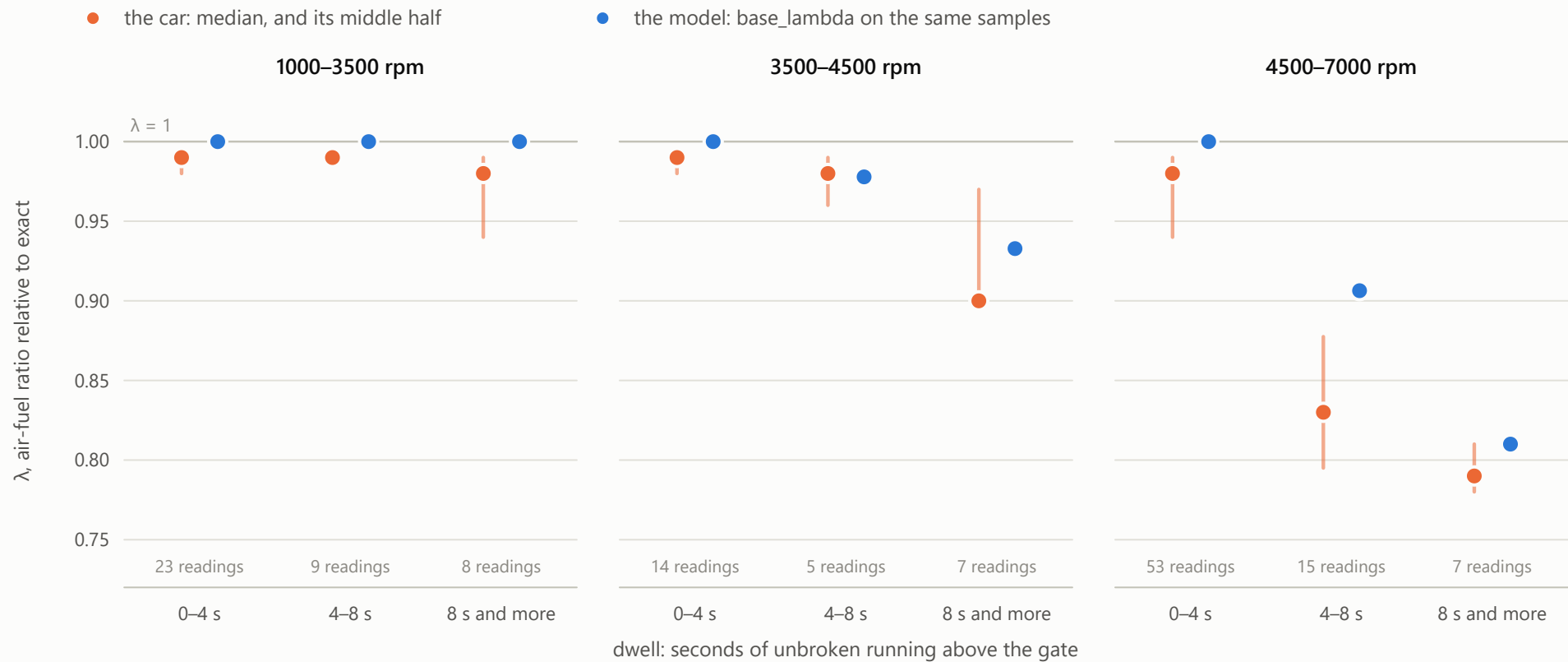


Grey: every steady sample with the air-mass sensor below its ceiling; darker means more samples. Orange: the 95th percentile of pressure ratio in each bin (fit\_envelope.py). Read the counts on the top bins: the rows are forward-filled, and the number beside each dot is how many INDEPENDENT readings it rests on (AUDIT.md H4). Beyond the last bin nothing is measured, because the air-mass sensor saturates (CLAUDE.md mistake 7). Data: data/master\_samples.csv.

## Enrichment: the mixture the car ran under boost, against the model's map

1,341 samples above the 180 kPa gate · nine cells by engine speed and seconds of dwell · largest gap 0.076 in  $\lambda$

**FIT** base\_lambda v4 was fitted to these same samples (engine\_env.BaselineECU); it tests nothing



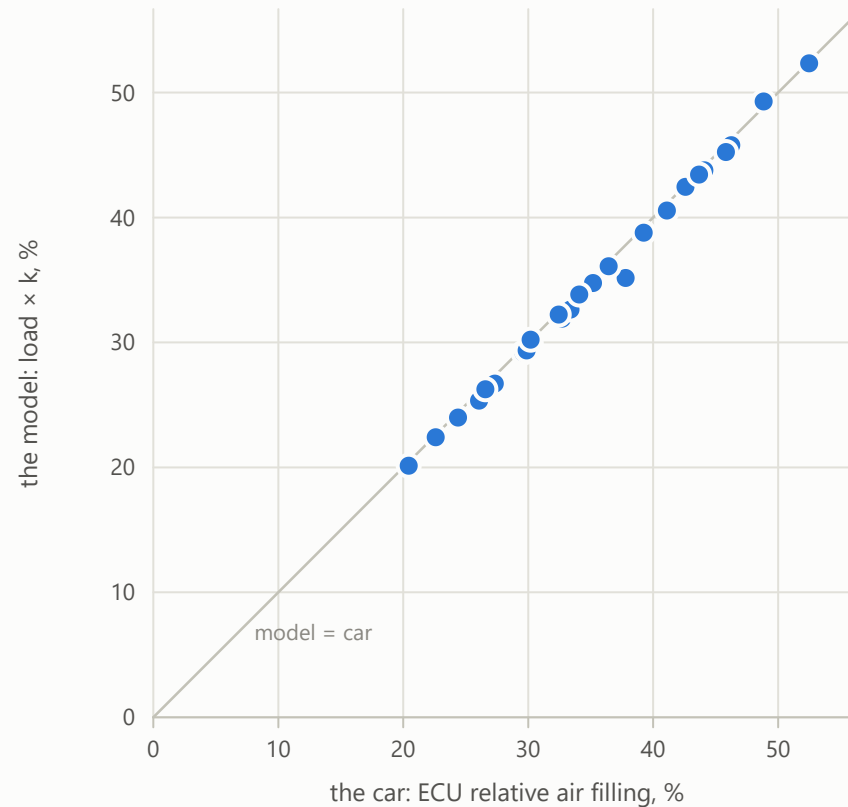
$\lambda = 1$  is the chemically exact mixture; below 1 is rich, fuel spent to cool the exhaust. Dwell = seconds of unbroken running above the gate, from the timestamps (verify\_docs.dwell\_column). Dots are medians over the cell; the orange bar is the middle half of the car's samples; the count under each cell is how many independent lambda readings it rests on (AUDIT.md H4). The map's dwell constants were fitted before AUDIT.md H3 corrected the dwell axis, and the worst cell shows it: 4500–7000 rpm at 4–8 s, car 0.83 against model 0.91. The locked scenario's climb runs in the first panel's band, where the map never enriches. Data: data/master\_samples.csv.

## Engine load at the steady points: the model against the ECU's own load channel

26 points, 30–75 kPa manifold pressure · mean error 1.4 % with nothing fitted · 1.1 % with one fitted constant

### CONSISTENCY CHECK

not a test of the breathing model:  $\eta_v$ , residual gas and intake temperature cancel out



Horizontal: the ECU's 'Relative air filling'. Vertical: the model's load from the measured air mass, times  $k = 269.6 / T_{\text{charge}}$  -- the DIN reference state, 1013 mbar and 0 °C -- with nothing fitted. Both sides are built from the same air-mass reading, so the agreement shows what the ECU channel means and that the model has the right displacement (a wrong engine misses badly); it cannot show that the model breathes like the car (CLAUDE.md mistake 12). 'Steady' means steady in road and engine speed only (AUDIT.md M3). Data: data/master\_points.csv.

# The numbers behind each chart

## 1. The gearbox: the ratios the car ran, against the eight Toyota publishes

| gear | published overall ratio | samples within $\pm 4\%$ |
|------|-------------------------|--------------------------|
| 1    | 16.537                  | 12                       |
| 2    | 10.584                  | 348                      |
| 3    | 6.842                   | 3,598                    |
| 4    | 5.418                   | 7,551                    |
| 5    | 4.145                   | 9,187                    |
| 6    | 3.150                   | 8,189                    |
| 7    | 2.589                   | 7,314                    |
| 8    | 2.016                   | 32,376                   |

## 2. Boost pressure under boost: the model against the car's own boost channel

| side                                   | samples | median, kPa | gap to the car |
|----------------------------------------|---------|-------------|----------------|
| the car: Boost pressure channel        | 1,097   | 224.0       | --             |
| the model: charge_temperature()        | 762     | 228.3       | +1.9 %         |
| the old error: the pre-throttle sensor | 587     | 279.5       | +24.8 %        |

## 3. Boost against air flow: the simulator's ceiling on the car's envelope

| flow bin, kg/s | car p95 PR | model PR | rows   | independent readings |
|----------------|------------|----------|--------|----------------------|
| 0.015          | 1.159      | 1.198    | 24,060 | 1,120                |
| 0.045          | 1.312      | 1.507    | 14,759 | 809                  |
| 0.075          | 1.582      | 1.735    | 1,040  | 68                   |
| 0.105          | 1.933      | 1.911    | 166    | 15                   |
| 0.135          | 2.283      | 2.050    | 249    | 22                   |

|       |       |       |     |    |
|-------|-------|-------|-----|----|
| 0.165 | 2.353 | 2.164 | 142 | 15 |
| 0.195 | 2.440 | 2.258 | 124 | 13 |
| 0.225 | 2.415 | 2.337 | 36  | 5  |
| 0.255 | 2.342 | 2.405 | 40  | 5  |
| 0.285 | 2.515 | 2.463 | 53  | 4  |
| 0.315 | 2.473 | 2.514 | 47  | 5  |

#### 4. Enrichment: the mixture under boost, against the model's map

| cell                                 | rows | independent readings | car median | car middle half | model |
|--------------------------------------|------|----------------------|------------|-----------------|-------|
| $\lambda$ cell: 1000–3500 rpm, 0–4 s | 249  | 23                   | 0.99       | 0.98 to 0.99    | 1.00  |
| $\lambda$ cell: 1000–3500 rpm, 4–8 s | 123  | 9                    | 0.99       | 0.99 to 0.99    | 1.00  |
| $\lambda$ cell: 1000–3500 rpm, 8+ s  | 69   | 8                    | 0.98       | 0.94 to 0.99    | 1.00  |
| $\lambda$ cell: 3500–4500 rpm, 0–4 s | 137  | 14                   | 0.99       | 0.98 to 0.99    | 1.00  |
| $\lambda$ cell: 3500–4500 rpm, 4–8 s | 45   | 5                    | 0.98       | 0.96 to 0.99    | 0.98  |
| $\lambda$ cell: 3500–4500 rpm, 8+ s  | 53   | 7                    | 0.90       | 0.90 to 0.97    | 0.93  |
| $\lambda$ cell: 4500–7000 rpm, 0–4 s | 435  | 53                   | 0.98       | 0.94 to 0.99    | 1.00  |
| $\lambda$ cell: 4500–7000 rpm, 4–8 s | 142  | 15                   | 0.83       | 0.80 to 0.88    | 0.91  |
| $\lambda$ cell: 4500–7000 rpm, 8+ s  | 88   | 7                    | 0.79       | 0.78 to 0.81    | 0.81  |

#### 5. Engine load at the steady points: the model against the ECU's load channel

| rpm  | manifold kPa | ECU load, % | model load $\times$ k, % | error, % |
|------|--------------|-------------|--------------------------|----------|
| 2439 | 30           | 20.4        | 20.1                     | -1.5     |
| 2931 | 33           | 22.6        | 22.4                     | -0.8     |
| 2582 | 36           | 24.4        | 24.0                     | -1.6     |
| 1552 | 40           | 26.1        | 25.3                     | -2.8     |
| 2443 | 40           | 26.6        | 26.3                     | -1.2     |
| 1879 | 42           | 27.3        | 26.7                     | -2.3     |
| 2039 | 45           | 29.6        | 29.2                     | -1.4     |
| 3178 | 44           | 29.9        | 29.4                     | -1.7     |
| 5053 | 45           | 30.2        | 30.2                     | +0.1     |

|      |    |      |      |      |
|------|----|------|------|------|
| 2615 | 44 | 30.2 | 29.8 | -1.4 |
| 4184 | 47 | 32.4 | 32.2 | -0.6 |
| 1553 | 50 | 32.7 | 31.9 | -2.5 |
| 1707 | 50 | 33.4 | 32.6 | -2.2 |
| 2935 | 49 | 34.1 | 33.8 | -0.7 |
| 1878 | 53 | 35.2 | 34.8 | -1.2 |
| 2615 | 53 | 36.4 | 36.1 | -0.9 |
| 1549 | 54 | 37.8 | 35.2 | -7.0 |
| 2039 | 58 | 39.2 | 38.8 | -1.2 |
| 1876 | 61 | 41.1 | 40.6 | -1.3 |
| 2614 | 61 | 42.6 | 42.5 | -0.3 |
| 2918 | 61 | 43.7 | 43.4 | -0.5 |
| 2882 | 62 | 44.1 | 43.8 | -0.7 |
| 2260 | 66 | 45.8 | 45.2 | -1.3 |
| 2040 | 67 | 46.3 | 45.8 | -1.0 |
| 1879 | 73 | 48.8 | 49.3 | +0.9 |
| 2274 | 75 | 52.5 | 52.3 | -0.3 |